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1 Metropolis-Hastings Monte Carlo

We want to generate a set of samples from a distribution $P(x)$. The goal is to use a Markov process that asymptotically reaches a distribution that coincides with $P(x)$.

We start with the detailed balanced condition: consider a system in thermal equilibrium, and two states x and x' of that system. The probability of being in state x and going from x to x' is the same as the probability of being at x' and going back to state x:

$$P(x \to x')P(x) = P(x' \to x)P(x'). \tag{1}$$

Here $P(x \to x')$ is occasionally written as $P(x' \mid x)$, as in a Markov process we can consider the probability of going to x' *given* a previous state x. This is fact would require us to factor $P(x' \mid x)$ into $P(x' \mid x) = g(x' \mid x)A(x', x)$, where $g(x' \mid x)$ captures the *conditional probability of considering x' as the next state* that might arise if this process is such that the probability distribution *changes* depending on the previous state, and $A(x', x)$ is the probability of actually making that transition. In statistical mechanics the equilibrium probability distribution does not change depending on which of the states of the ensemble you're in, so the above is not really a point of concern and we can unambiguously write $P(x \to x')$.

The idea is to write eq. (1) as

$$\frac{P(x \to x')}{P(x' \to x)} = \frac{P(x')}{P(x)} \tag{2}$$

and to choose $P(x \to x')$ and $P(x' \to x)$ such that this condition is always fulfilled. An intuitive choice is the so-called *Metropolis* choice, which is the most important bit to understand:

$$P(x \to x') = \min\left(1, \frac{P(x')}{P(x)}\right)$$

Make the move from x to x' no matter what if $P(x') > P(x)$ (i.e., if x' is a more likely state); make the move with probability $P(x')/P(x)$ if it is a less likely state.

Where in the algorithm we generate a random number between 0 and 1 and make the move if that number is less than $P(x \to x')$ (hence with certainty if $P(x') > P(x)$ and with chance to $P(x')/P(x)$ otherwise). Here implicitly the same choice is also made for $P(x' \to x)$, where we just switch $x \longleftrightarrow x'$ in the above definition (note that we only have to compute $P(x \to x')$, though, since we are only considering the move from x to x'). This fulfilled eq. (2) in the following sense:

- If $P(x') > P(x)$, we set $P(x \to x') = 1$ and $P(x' \to x) = P(x)/P(x')$. Then the ratio of $P(x \to x')$ and $P(x' \to x)$ is the right hand side of eq. (2):

$$\frac{P(x \to x')}{P(x' \to x)} = \frac{1}{P(x)/P(x')} = \frac{P(x')}{P(x)}$$

- If $P(x') < P(x)$, we set $P(x \to x') = P(x')/P(x)$ and $P(x' \to x) = 1$ and again the ratio eq. (2) is satisfied.

The power in its application to statistical mechanics lies in the fact that we only need to compute the relative change in energy $P(x')/P(x)$! E.g., in the Ising model with coordination number z, if we are in an all-up (or down) ferromagnetic state and flip spin k:

$$\frac{P(x')}{P(x)} = \frac{\exp\left(\beta J \sum S_i S_j\right)}{\exp\left(\beta J \sum S_i S_j\right)} = \frac{\cancel{\exp\left(\beta J \sum_{i,j\neq k} S_i S_j\right)} \cdot \exp\left(\beta J \sum_j S_j S_k\right)}{\cancel{\exp\left(\beta J \sum_{i,j\neq k} S_i S_j\right)} \cdot \exp\left(\beta J \sum_j S_j S_k\right)}$$
$$= \frac{\exp\left(\beta J \sum_j (+1)(-1)\right)}{\exp\left(\beta J \sum_j (+1)(+1)\right)} = \frac{\exp\left(-z\beta J\right)}{\exp\left(z\beta J\right)} = \exp(-2zJ\beta).$$

2 Symmetries and Degeneracy

We use here the example of rotational symmetry, although the argument is general unless otherwise specified. Our goal is essentially to derive the m (magnetic quantum number) degeneracy present in central potentials/rotationally invariant systems such as the hydrogen atom. Consider the $(2\ell+1)$-dimensional irrep of L_z, the angular momentum operator. We know this generates the group of rotations about the z axis by $\exp(-i\theta L_z)$. Consider also a Hamiltonian with z-rotational symmetry. Then

$$e^{-i\theta L_z} H e^{i\theta L_z} = H \implies \left[e^{i\theta L_z}, H\right] = 0$$

We can ascertain from this two facts:

1. z-rotated states are also eigenstates of H with the same energy: $\left[e^{i\theta L_z}, H\right] = 0 \implies H e^{i\theta L_z} |E_n\rangle = e^{i\theta L_z} H |E_n\rangle$, so $H(e^{i\theta L_z} |E_n\rangle) = E_n(e^{i\theta L_z} |E_n\rangle)$. In a sense this can be thought of as a degeneracy, but this is not a particularly useful line of thinking: applying the rotation operator $e^{i\theta L_z}$ does not alter the quantum numbers of E_n with respect to the quantization axis L_z, so there are no other eigenstates of L_z with the same energy, and this "degeneracy" is fictitious, so to speak. Besides, $e^{i\theta L_z}$ is not Hermitian and thus is not an observable.

2. Eigenstates of H are also eigenstates of L_z: $\left[e^{i\theta L_z}, H\right] = 0 \implies [L_z, H] = 0$. Note this does not mean that the L_z-eigenvalues are the same as H eigenvalues! That is, we've not yet derived the degeneracy; in fact, if the eigenvalues were the same, we might've indeed derived a degenerate set of states. What this does mean is that we can use both E_n and m, ℓ to label the eigenstates.

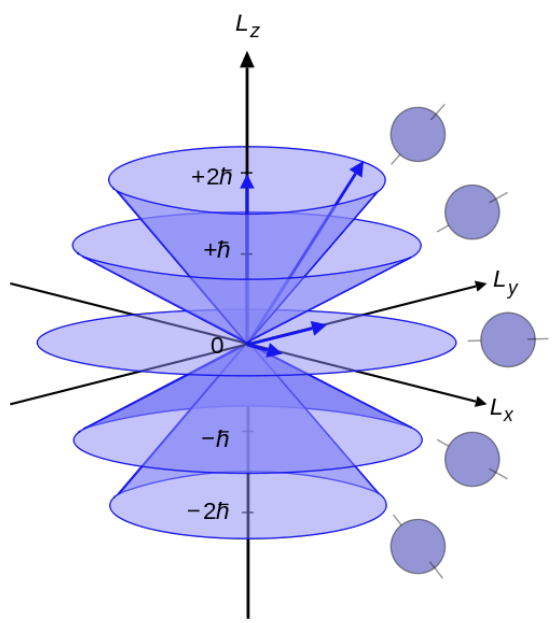

Figure 1

In reference to the latter fact, I will only use m henceforth, dropping the quantum number ℓ. This amounts to implicitly picking the irreducible representation of dimension $2\ell+1$ of the rotation group.

So far we've only considered z-rotational symmetry. This does not give us the degeneracy in m. To do this, we need some more physical intuition: we know that the magnetic quantum number m roughly corresponds to the projection of the angular momentum along the z axis. Degeneracy in m corresponds to the Hamiltonian produce *the same energies independent of the direction of the angular momentum* (specifically, its projection along the z axis). To obtain this, the system needs to also be invariant with respect to rotations along the x- and y- axes, as such rotations will alter the projection along the z-axis. See fig. 1. This is the main insight:

Symmetries lead to Hamiltonians that have solutions with the same energy related by a certain transformation. If that transformation changes a certain quantum number, it should be the case that those new states (labelled by the new quantum numbers) are also solutions with the same energies, and thus those transformations generate the degeneracy in that quantum number.

Then the generator L_x should correspond to the transformations that lead to the degeneracy. This is seen by considering the transformation law $L_x = \frac{1}{2}(L_+ + L_-)$ (again, if m is the projection along the z axis, we can see that rotations $e^{i\theta L_x}$ increase, and, as a superposition, increase the projection of the angular momentum along the z axis, exactly as if we are rotating the angular momentum vector). Then, if the Hamiltonian is symmetric with respect to x-rotations,

$$\begin{aligned}
L_x H \left|E_n, m\right\rangle &= H L_x \left|E_n, m\right\rangle \\
\frac{1}{2} E_n (L_+ + L_-) \left|E_n, m\right\rangle &= \frac{1}{2} H (L_+ + L_-) \left|E_n, m\right\rangle \\
\frac{1}{2} E_n L_\pm \left|E_n, m\right\rangle &= \frac{1}{2} H L_\pm \left|E_n, m\right\rangle && (\left|m \pm 1\right\rangle \text{ are linearly indep.}) \\
E_n \left|E_n, m \pm 1\right\rangle &= H \left|E_n, m \pm 1\right\rangle .
\end{aligned}$$

Thus altering the value of m leads to another eigenstate with the same energy.

Some other remarks: its possible that the symmetry group we are considering is Abelian. In this case, it turns out that the irreducible representation is trivial (see here). This means that for Abelian symmetries, the state is only altered by an overall phase factor and there is no degeneracy: if $\left[H, e^{it\theta}\right] = 0$

$$E_n e^{it\phi} \left|E_n\right\rangle = H e^{it\phi} \left|E_n\right\rangle \Longrightarrow e^{it\phi} E_n \left|E_n\right\rangle = e^{it\phi} H \left|E_n\right\rangle \Longrightarrow E_n \left|E_n\right\rangle = H \left|E_n\right\rangle$$

where ϕ is the one dimensional representation. The last equality, of course is something we already knew.

An example is that of a one-dimensional symmetric potential; the symmetry is $\mathbb{Z}_2$, which is Abelian. Such a system has no degeneracy.

LRL vector and ℓ degeneracy

Inverse-square potentials are known to have an additional conserved quantity, the Laplace-Runge-Lenz vector. This turns out to correspond to a 4-dimensional spherical symmetry, which in turn leads to a degeneracy in the ℓ (azimuthal) quantum numbers, such as those present in the Hydrogen atom.

See also

- The examples section of this document.

3 Connections, Generally

There is a connection between connections (haha); that is, when we hear people refer to the potentials A_μ as "connections" in gauge theory (quantum field theory), this actually does correspond to the Riemannian connection $\nabla_X Y$ we see in differential geometry, and similarly for notions such as curvature, covariant derivatives, etc.

There is one quick terminological clarification to make. What we refer to as a connection in gauge theory (e.g., A_μ) really corresponds to the Christoffel symbols Γ^k_{ij}; the (gauge) covariant derivative is, under this terminology, really $\nabla_X Y$. The reason the two are referred to interchangeably is that the Γ^k_{ij} can be thought of as implicitly

determined upon specifying $\nabla_X Y$, and vice-versa. It's not unlike how people will refer to the representation space as a "representation." Which infuriates me to this day.

One discrepancy that might be concerning at first is the fact that, in gauge theory (let's generalize to non-Abelian here), the connection appears to have one index: e.g., in Peskin and Schroeder §15.2 we see A^i_μ, where i indexes the generator of the gauge symmetry group. From Stack Exchange:

> Mathematicians would refer to a gauge theory setting as a fiber bundle. A_μ takes the role of a connection on the fibre bundle. Of course, out of the three indices the Cristoffel symbols Γ^i_{jk} have, two indices "live" in the fiber. You can appreciate this more easily if you look at non-Abelian gauge theories, in which $(A_\mu)_{ij} = A^a_\mu T^a_{ij}$

So here $(A_\mu)_{ij}$ corresponds to Γ^i_{jk}.

To understand the correspondence between curvatures, we need an expression of the curvature of Riemannian geometry $R(X,Y)Z$ in terms of the connection Γ^k_{ij}. The bottom equation of page 92 of Do Carmo's *Riemannian Geometry* gives us what we need:

$$R(X_i, X_j)X_k = \nabla_{X_j}(\Gamma^\ell_{ik}X_\ell) - \nabla_{X_i}(\Gamma^\ell_{jk}X_\ell). \tag{3}$$

(X_i are basis vectors, so it's not difficult to convert to index notation from this.)

- **Berry's phase:** The connection is the *Berry connection* $A^n(\mathbf{R}) = i\langle n(\mathbf{R})|\partial_{\mathbf{R}}|n(\mathbf{R})\rangle$ (**what is the fiber bundle?**), from which we obtain the *Berry curvature* $\omega_{\mu\nu}(\mathbf{R}) = \partial_{R_\mu}A^n_\nu(\mathbf{R}) - \partial_{R_\nu}A^n_\mu(\mathbf{R})$. Cf. eq. (3); the missing index relative to $R(X_i, X_j)X_k$ comes from the fact that we've written $\mathbf{R}$ in the vector form as opposed to index form.

See also

- Girvin and Yang's *Modern Condensed Matter Physics*, §13.1 has an excellent explanation of Berry's phase.
- These notes.

4 Legendre Transforms

This is one I've been meaning to sit down and understand more carefully for one or two years, actually. It's a remarkably simple concept that somehow gets mangled by thoughtless pedagogy in almost every single treatment. The big idea is: we'd like to turn a function $f(x)$ into a function of $p = \mathrm{d}f/\mathrm{d}x$. Put in other words, we can think of it as "a map from the graph of the function ot the tangents of the graph" (Wikipedia).

Say we have a convex function $f(x)$ (in the case the function isn't convex everywhere, we can restrict ourselves to an interval where it is convex), and a line $y = px + b$ which we'd like to make the tangent line of $f(x)$ at $x = x_0$ by tweaking p and b. For this to be the case,

$$y = px + b = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0} x + b \Longrightarrow p = \left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_{x_0}$$
$$\text{and} \qquad y(x = x_0) = px_0 + b = f(x_0)$$

(Note this is *not* related to the Taylor series expansion, we're just looking for the tangent line at the point $f(x_0)$). Since $f(x)$ is convex $\mathrm{d}f/\mathrm{d}x = p$ can be inverted, and we can solve for $x_0 = (f')^{-1}(p)$. Now it remains to solve for b by using the last equality above:

$$b = f(x_0) - px_0$$

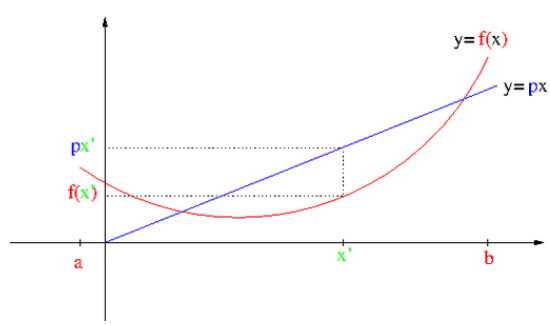

Figure 2

Since we know $x_0 = (f')^{-1}(p)$ and we're ultimately looking to convert f into a function of p:

$$b = f((f')^{-1}(p)) - p \cdot (f')^{-1}(p) \equiv f^*(p)$$

were we've defined the Legendre transform in the last equality. As it turns out, the value of x_0 is such that the distance between $f(x_0)$ and px_0 is maximized: we could have found x_0 as a function of p by maximizing $b = b(x)$:

$$\begin{aligned} b(x) = f(x) - px \Longrightarrow \left.\frac{\partial b}{\partial x}\right|_{x_0} &= f'(x_0) - p = 0 \\ \Longrightarrow x_0 &= (f')^{-1}(p), \end{aligned}$$

which is what commonly drawn figures such as fig. 2 are trying to tell you.

> The Legendre transform is defined by
>
> $$f^*(p) = f(x_0) - px_0, \qquad x_0 = (f')^{-1}(p),$$
>
> It states that for a convex function f we can determine the function's tangent lines' y-intercepts for given values p of their slopes. $f^*(p)$ is precisely the y-intercept of the function for a slope p.

Here's an interactive Desmos graph that demonstrates this. In this sense it is a very simple procedure—one a high school calculus student could discover by accident if, say, they are trying to make an animation of a tangent line coasting along the plot of a function.

This revisit was informed by an issue Elijah brought up. It was something like: one encounters a "path integral" with "Lagrangian" $\pi\phi - H[\pi, \phi]$ integrated over both the field *and* the canonical momentum π; this it *not* the usual path integral, as there is a mysterious integration over π. We were curious as to why. The answer is that the the Legendre transform exchanges

$$H(p, q) \longleftrightarrow L(\dot{q}, q), \qquad \text{i.e.,} \qquad p \longleftrightarrow \dot{q},$$

where $p \equiv \partial L/\partial \dot{q}$ *is evaluated at a point specified by* q, t, so properly it should be written $\partial L/\partial q\,|_{\dot{q}_0, q, t}$. This is why the path integral resulting from the proper Legendre transform only includes integration over the field ϕ, π is implicitly given by ϕ.